

Course: Cloud and Network Security CNS-C2-2026

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Week 1 Assignment 2:

Lab – use Wireshark to view Network Traffic

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Objectives

Part 1: Capture and analyse local icmp data in Wireshark

Part 2: Capture and analyse remote icmp data in Wireshark

Introduction

This lab uses Wireshark, a protocol analyser ("packet sniffer"), to capture and examine live network traffic. Wireshark captures each protocol data unit (PDU) crossing the network and decodes it against the relevant RFC or specification, making it a valuable tool for troubleshooting, analysis, and protocol development. In this exercise, Wireshark is used to capture ICMP traffic and identify the IP addresses and Ethernet MAC addresses involved in a ping exchange.

Part 1: Capture and Analyse Local ICMP Data in Wireshark

Step 1 Retrieve pc interface address

The screenshot shows a Windows desktop environment. In the background, a Google search for 'Wireshark' is visible. In the foreground, there are three main windows:

- Wireshark:** The interface shows the 'Packet List' pane with a single entry for an ICMP Echo (ping) request. The 'Packet Details' pane shows the structure of the ICMP Echo request, including the type, code, and checksum. The 'Packet Bytes' pane shows the raw data of the packet.
- Command Prompt:** The window displays the output of the 'ipconfig /all' command, showing the network configuration for the 'Ethernet adapter Ethernet' and 'Wireless LAN adapter Wi-Fi'. The Ethernet adapter is configured with the IP address 192.168.100.1, subnet mask 255.255.255.0, and default gateway 192.168.100.1. The Wireless LAN adapter is configured with the IP address 192.168.100.1, subnet mask 255.255.255.0, and default gateway 192.168.100.1.
- Document1 - Word (Product Act...):** The document is open to page 2 of 2, showing the 'Completion requirements:' section. It states that the user must 'receive a grade' and 'receive a passing grade'.

Figure 1 pc config

Step 2: Start Wireshark and begin capturing data

Wifi

packets

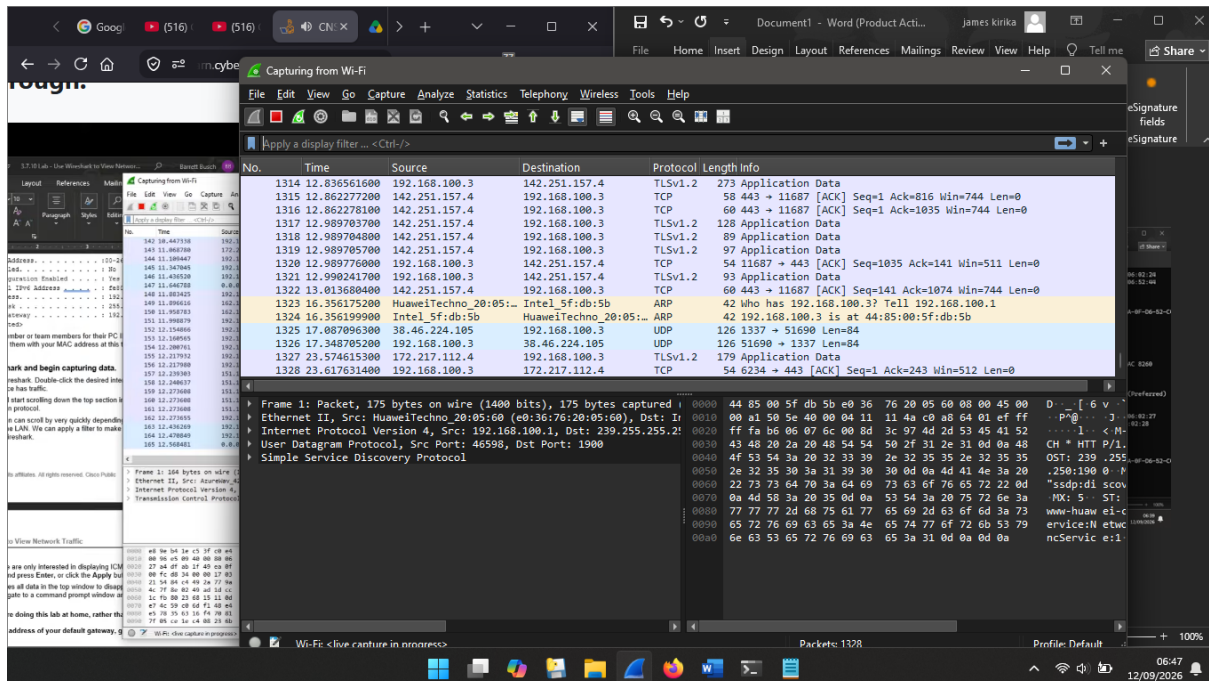


Figure 2 packets

Apply icmp filter and ping my phone ip address

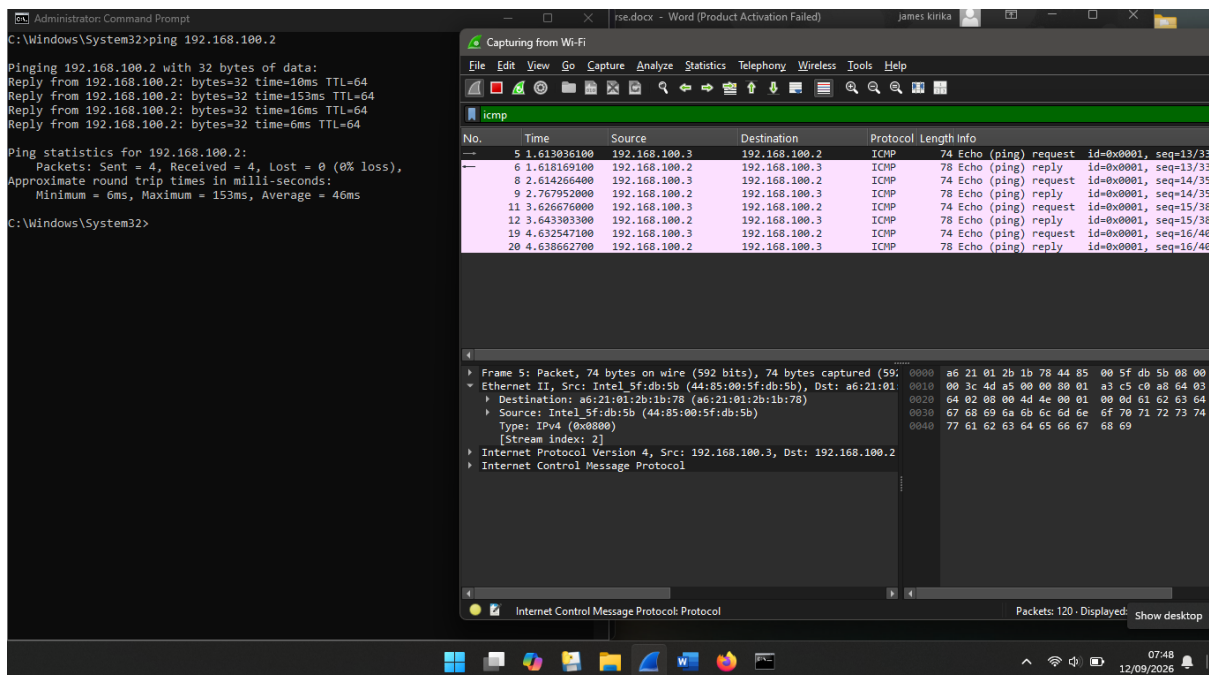


Figure 3 packet icmp filtered

Step 3: Examine the captured data.

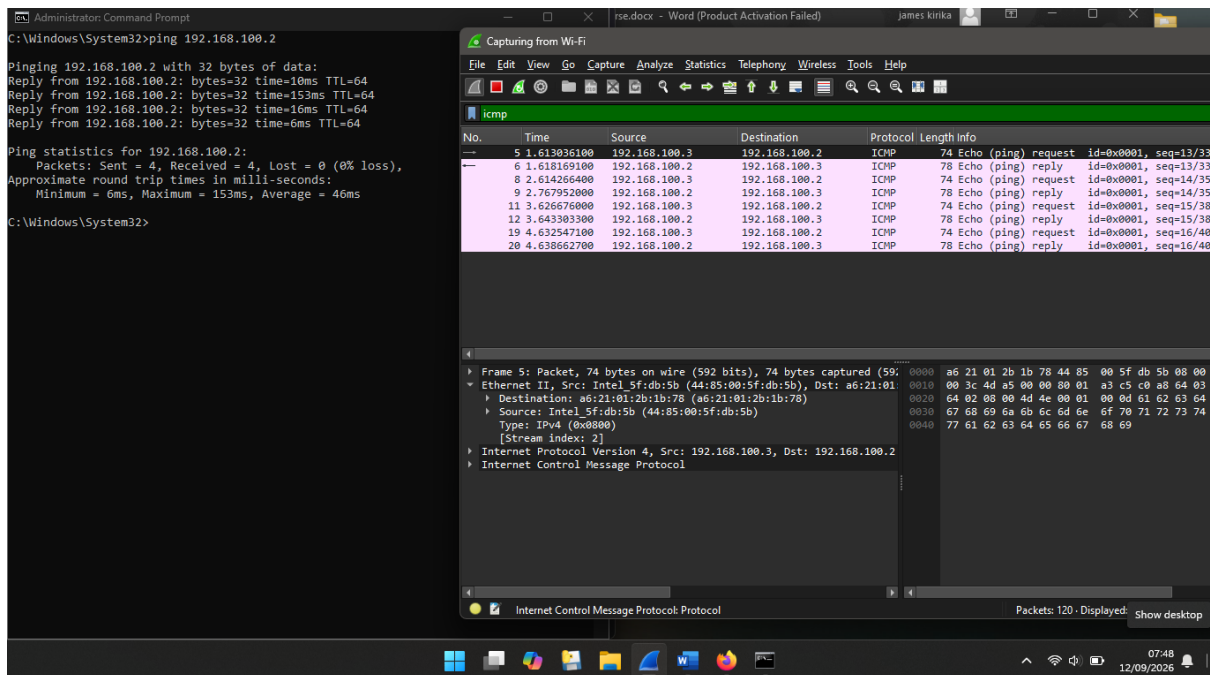


Figure 4 captured data

Question:

Does the source MAC address match pc interface? Yes.

Does the destination MAC match your team member? Yes

How is the MAC address of the pinged pc obtained by your pc? My pc broadcasts arp request asking who is in a particular IP Then the device with the said ip respond "it's me". Review Figure 5 arp filter: the image shows arp request broadcast, my pc (Intel in step no 3) is asking who is on 192.168.100.2 and my phone in (step no 4) confirms and shares its MAC address.

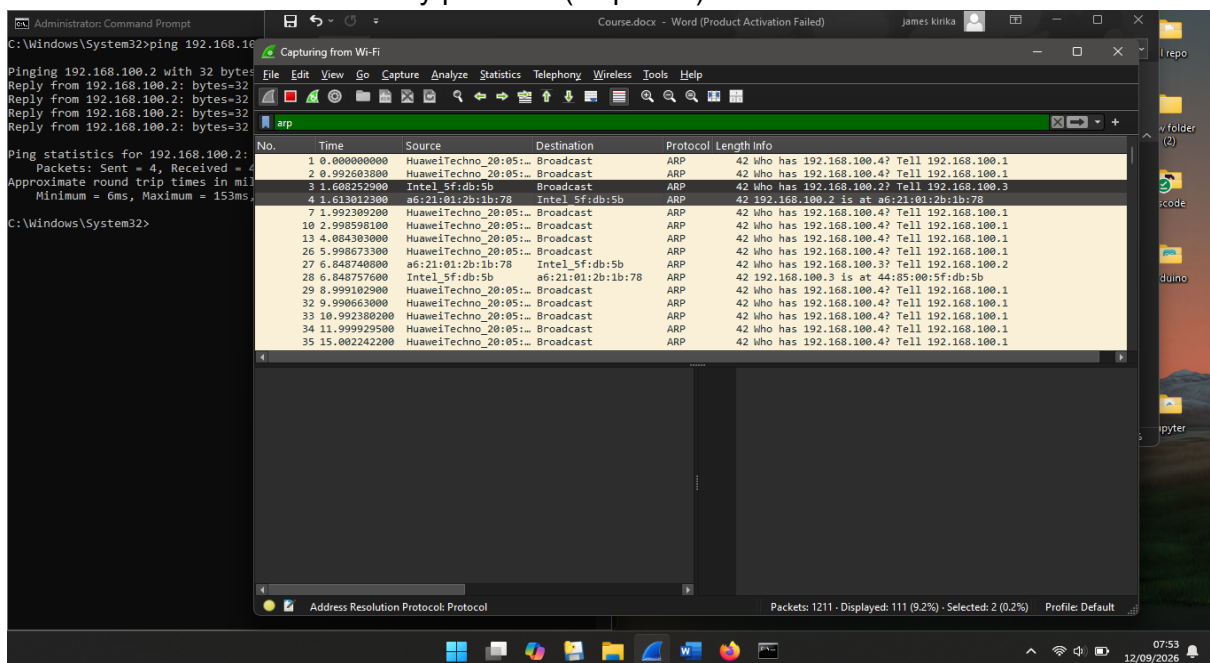


Figure 5 arp filter

Part 2: Capture and Analyse Remote ICMP Data in Wireshark

Step 1: Start capturing data on interface

Yahoo ping

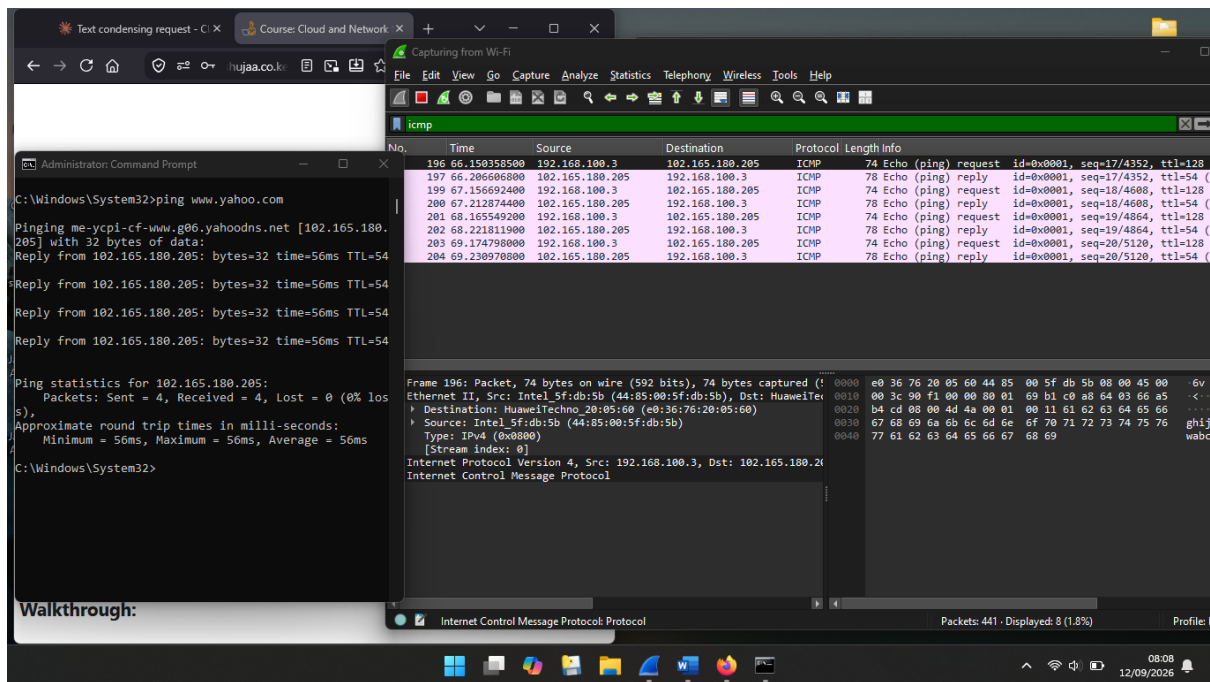


Figure 6 yahoo ping

Question

Ip address for www.yahoo.com: 102.165.180.205

Mac address for www.yahoo.com: Default Gateway router(e0:36:76:20:05:60)

Cisco ping

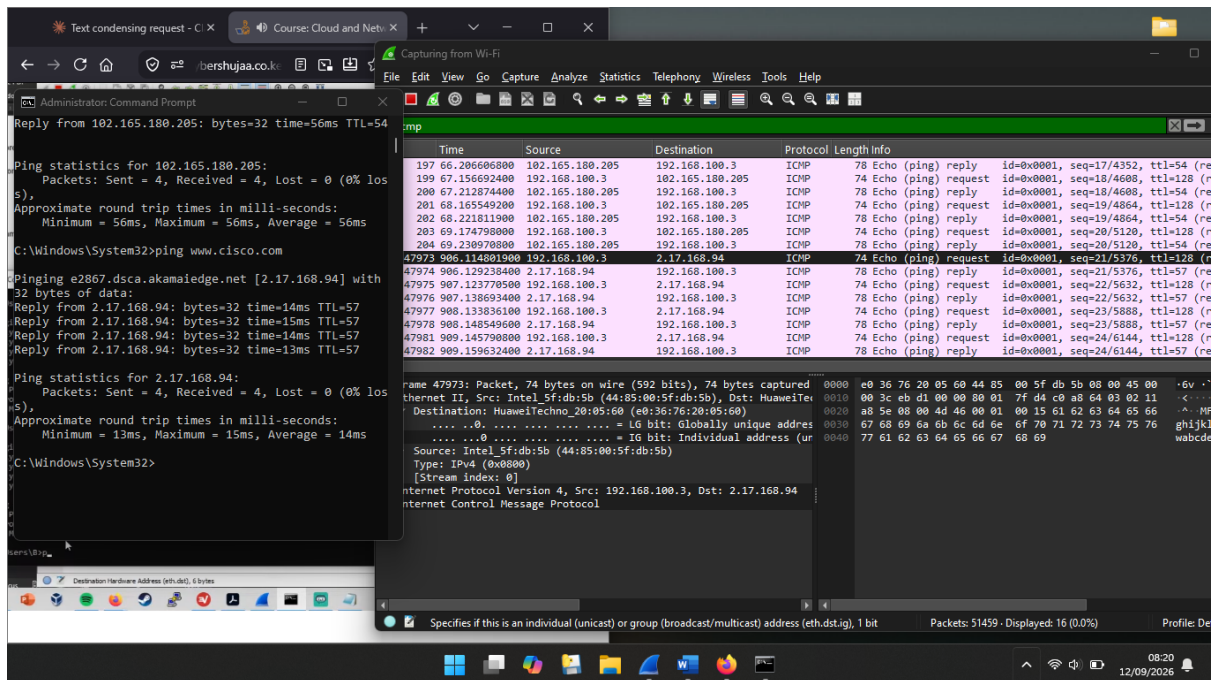


Figure 7 cisco ping

Question

Ip address for `www.cisco.com`: 2.17.168.94

Mac address for `www.cisco.com`: Default Gateway router(e0:36:76:20:05:60)

Google ping

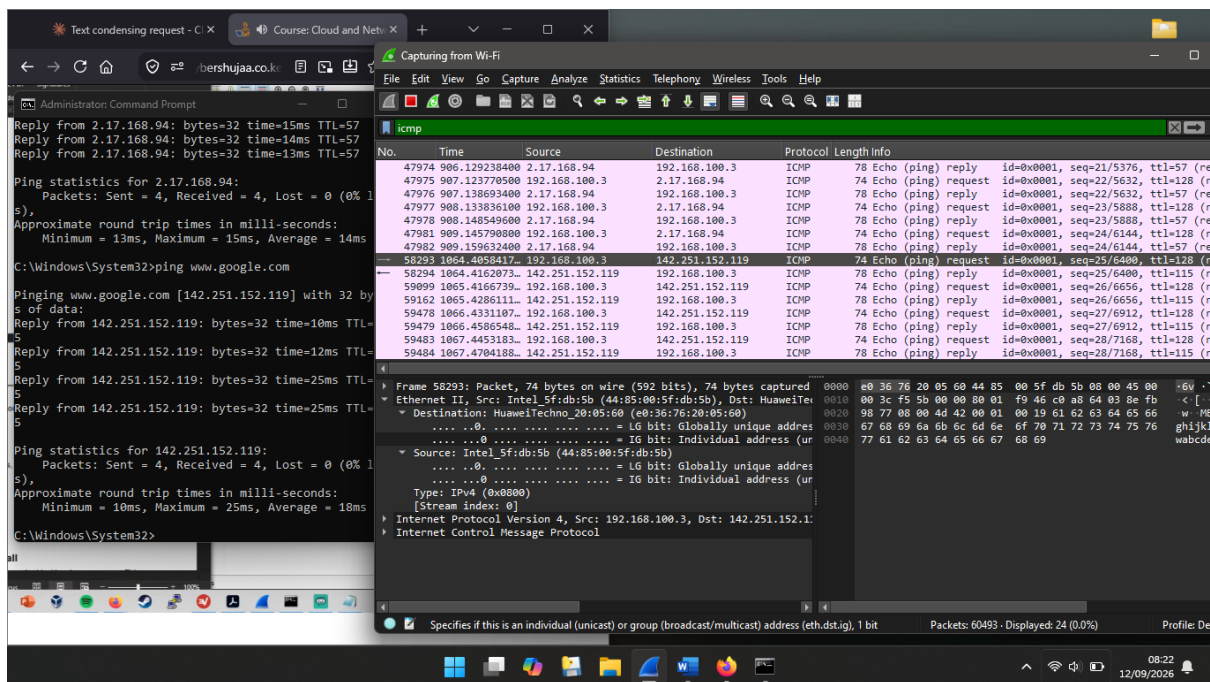


Figure 8 google ping

Question

Ip address for www.google.com: 142.251.152.119

Mac address for www.google.com: Default Gateway router(e0:36:76:20:05:60)

Reflection

MAC addresses only work on local network segment; they don't carry across the internet.

When you ping google.com, PC doesn't ARP for Google's MAC since the ip is not on the local subnet. Instead, it sends the frame to your **default gateway** (the router), because that's the only device it can reach directly on Layer 2.

Conclusion

This lab demonstrated how Wireshark can be used to capture and analyze live ICMP traffic at both the local and remote levels. In Part 1, examining a local ping confirmed that ARP resolves IP addresses to MAC addresses on the same subnet. MAC addresses matched the actual devices involved. In Part 2, pinging remote hosts (Yahoo, Cisco, and Google) showed that while the destination IP address reflected the actual remote server, the destination MAC address remained the default gateway's, since Layer 2 addressing only applies within the local network segment, and the router handles forwarding beyond it. Overall, the exercise reinforced the distinction between Layer 2 (MAC) and Layer 3 (IP) addressing, and clarified how traffic is routed from a local device out to hosts across the internet.